

Customer Shopping Behavior Analysis

Uncovering insights from 3,900 transactions to guide strategic business decisions.



Dataset at a Glance

3,900

Transactions analyzed

18

Feature columns

37

Missing values in Review Rating

Demographics

- Age, Gender, Location
- Subscription Status

Purchase Details

- Category, Amount, Season
- Size, Color, Shipping Type

Behavior

- Discounts, Promo Codes
- Frequency, Review Rating

Data Preparation in Python

01

Load & Explore

Imported with pandas; used `df.info()` and `.describe()`

02

Handle Missing Data

Imputed Review Rating nulls with per-category median

03

Standardize & Engineer

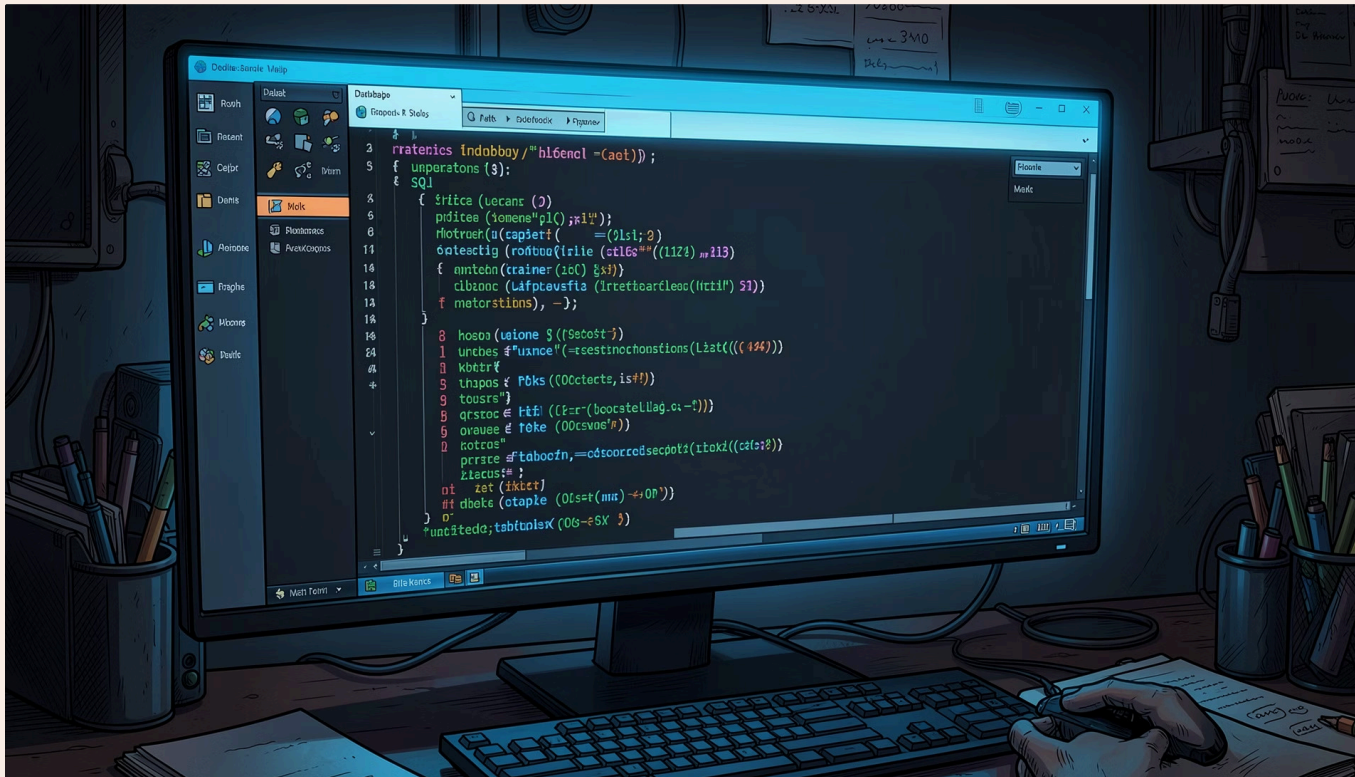
Snake_case columns; created `age_group` and `purchase_frequency_days`

04

Clean & Export

Dropped redundant `promo_code_used`; loaded to PostgreSQL

SQL Analysis: Revenue Insights



Revenue by Gender

Compared total revenue from male vs. female customers.

High-Spending Discount Users

Customers using discounts who still spent above average.

Shipping Type Comparison

Average spend: Standard vs. Express shipping.

Subscribers vs. Non-Subscribers

Subscribers

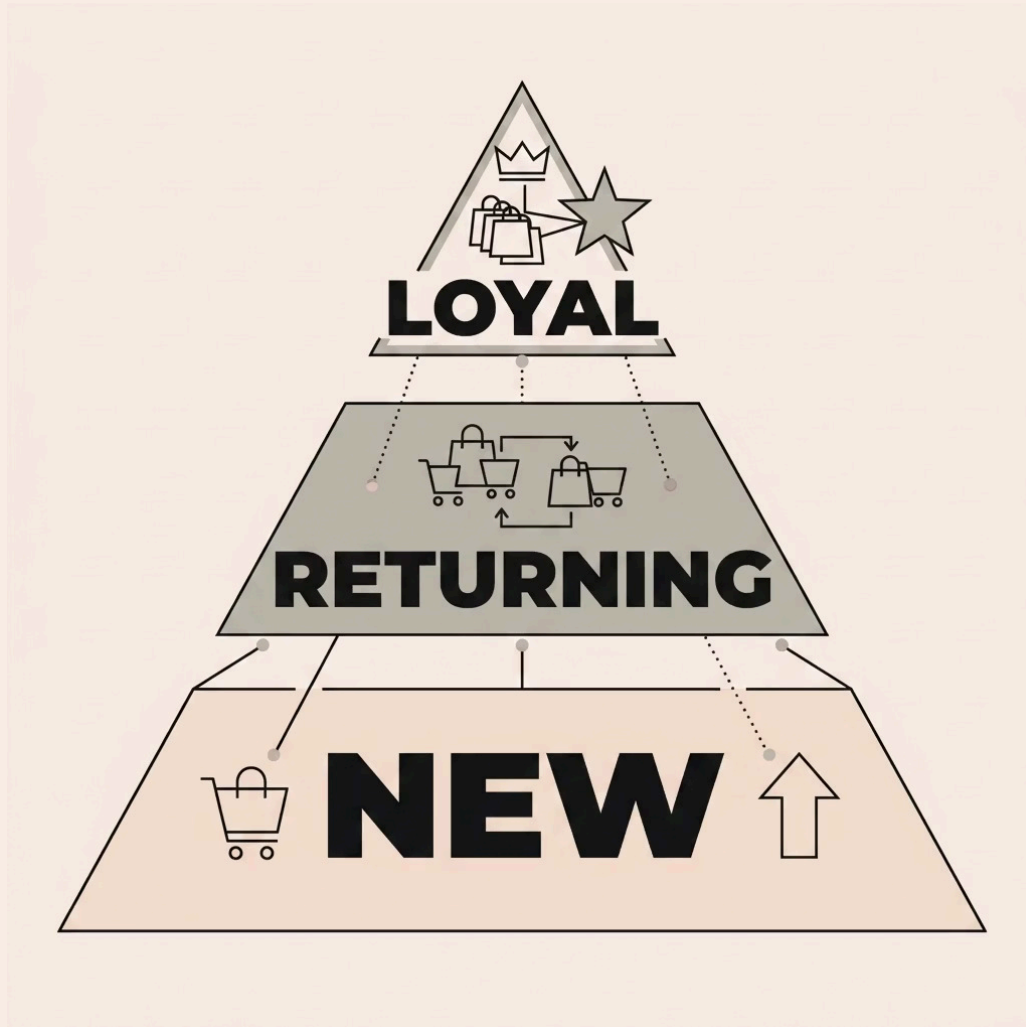
Higher average spend & total revenue contribution

Non-Subscribers

Lower engagement; greater discount dependency

 Repeat buyers with **>5 purchases** are significantly more likely to subscribe.

Customer Segmentation



Segment Logic

Classified by **previous purchase count** from transactional history.

- **New** – first-time buyers
- **Returning** – moderate history
- **Loyal** – high-frequency repeat buyers

Product & Category Insights



Top 5 by Rating

Products with highest average review scores identified



Discount-Dependent

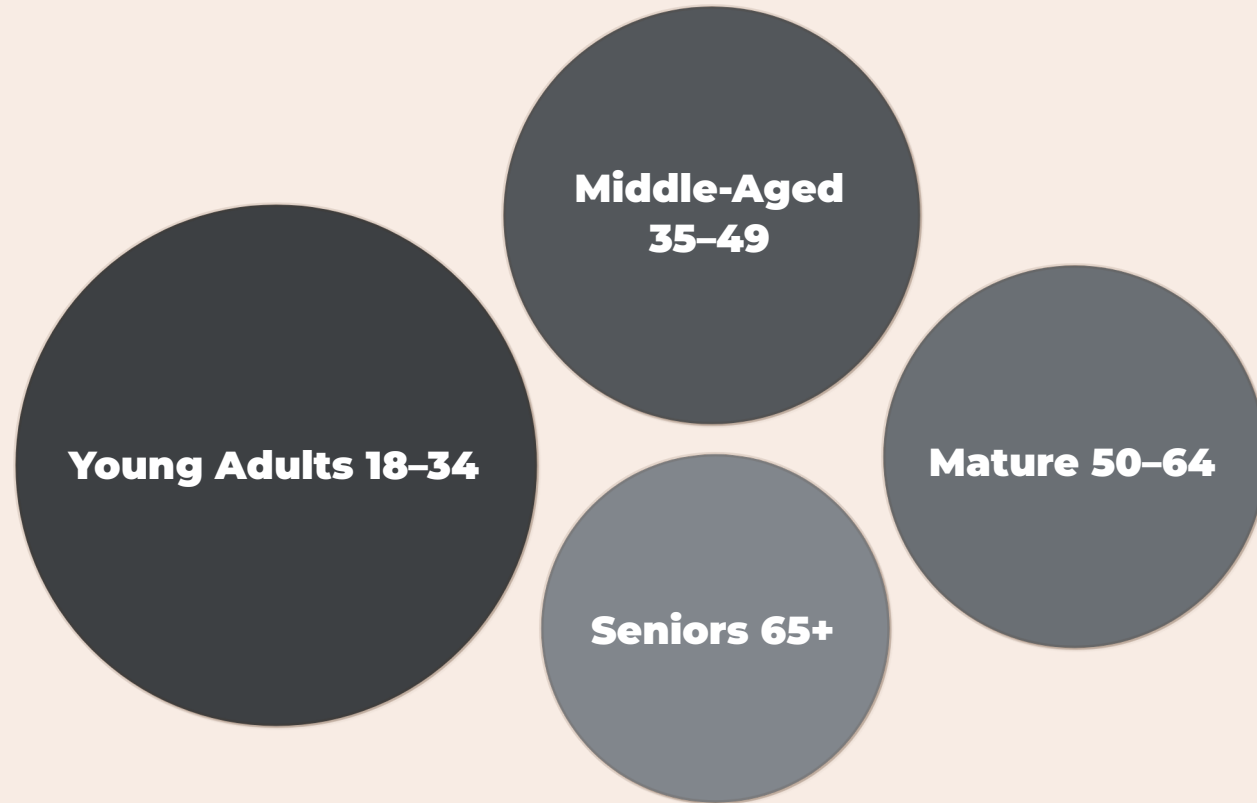
5 products with highest % of discounted purchases



Top 3 per Category

Most purchased products ranked within each category

Revenue by Age Group



Key Takeaway

Certain age bands drive disproportionate revenue – critical for **targeted marketing** spend allocation.

Power BI Dashboard

Interactive visuals presenting all key insights – segments, revenue, products, and behavior – in one unified view.



Business Recommendations

Boost Subscriptions

Promote exclusive subscriber benefits

Loyalty Programs

Reward repeat buyers; move them to "Loyal" segment

Review Discount Policy

Balance sales boosts with margin control

Product Positioning

Highlight top-rated & best-selling items in campaigns

Targeted Marketing

Focus on high-revenue age groups & express-shipping users